

Hospital Operations Analytics — Executive Summary

Prepared for: Hospital Operations Leadership **Data:** 1,000 patient records (Jan 2023 – Jan 2024) **Tools:** SQL (MySQL) for validation and KPI computation, Power BI for visualization

Overview

This analysis covers patient-level operational data across 6 departments, 7 diagnoses, 4 regions, and 3 insurance types. Headline metrics: **1,000 patients**, average treatment cost of **₹98,303**, an average length of stay of **8.15 days**, and a **50.2% readmission rate**.

Note on data: this dataset appears synthetically generated rather than drawn from real clinical records — for example, patient age is evenly distributed across every department regardless of specialty (e.g., Pediatrics spans ages 18–85). Findings below should be read as demonstrations of the analysis method rather than real-world clinical conclusions.

Key Findings

1. Orthopedics carries the highest average treatment cost

Average cost per Orthopedics patient is **₹1,04,757**, roughly 17% higher than the lowest-cost department, Neurology (**₹89,174**). Across all departments, cost varies by department more than it varies by region or insurance type — department is the strongest cost driver in this dataset.

Recommendation: Review Orthopedics case mix and resource allocation first when looking for cost-efficiency opportunities, since it has both the highest average cost and a meaningful patient volume.

2. Migraine patients have the highest readmission rate — and it compounds with cost

Migraine cases show a **57.1% readmission rate**, the highest of any diagnosis, well above Hypertension's **46.0%** (the lowest). The cost-vs-readmission view shows Migraine sitting in the high-cost, high-readmission quadrant relative to other diagnoses — making it a compounding risk rather than an isolated one.

3. Government-insured patients incur the highest average cost

Patients with Government insurance average **₹1,00,585** per treatment vs. **₹94,432** for Self-Pay patients — roughly a 6.5% gap. Patient satisfaction, by contrast, is nearly flat

across regions and outcomes (2.95–3.06 on a 5-point scale), suggesting satisfaction in this dataset is not strongly linked to cost, region, or treatment outcome.

Recommendation: Investigate whether the cost gap by insurance type reflects differences in case complexity, billing practice, or length of stay by payer — this is worth a follow-up analysis before drawing conclusions about cost drivers.

Data Quality Notes (from SQL validation)

Before trusting the KPIs above, the underlying data was validated in SQL:

- No duplicate patient records (1,000 total, 1,000 distinct PatientID)
 - No invalid date sequences (0 cases of discharge date preceding admission date)
 - Admission/discharge dates required explicit format handling (DD-MM-YYYY stored as text) — confirmed and corrected before computing length-of-stay metrics
 - Age/department/diagnosis independence flagged as a synthetic-data indicator (see Overview note above)
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Methodology

1. Data loaded into MySQL; validated for duplicates, invalid dates, and referential consistency
2. KPIs computed independently in SQL (see `healthcare_analysis.sql`) before building the BI layer, so dashboard numbers could be cross-checked against a second, independent calculation
3. Power BI dashboard built with 5 headline KPI cards, 5 supporting visuals, and 4 interactive filters (Region, Department, Insurance Type, Outcome) for live drill-down